CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau

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Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	41
Stimuli	1
Observed cells	1714
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96
Cell types	Extraembryonic_Endoderm, Neuroectoderm, Pluripotency, Post-implantation_Epiblast
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic); genes z-scored on the observed cells (clipped at ±10), PCA (30 components), 20 neighbours

Perturbations (perturbation_simulation)

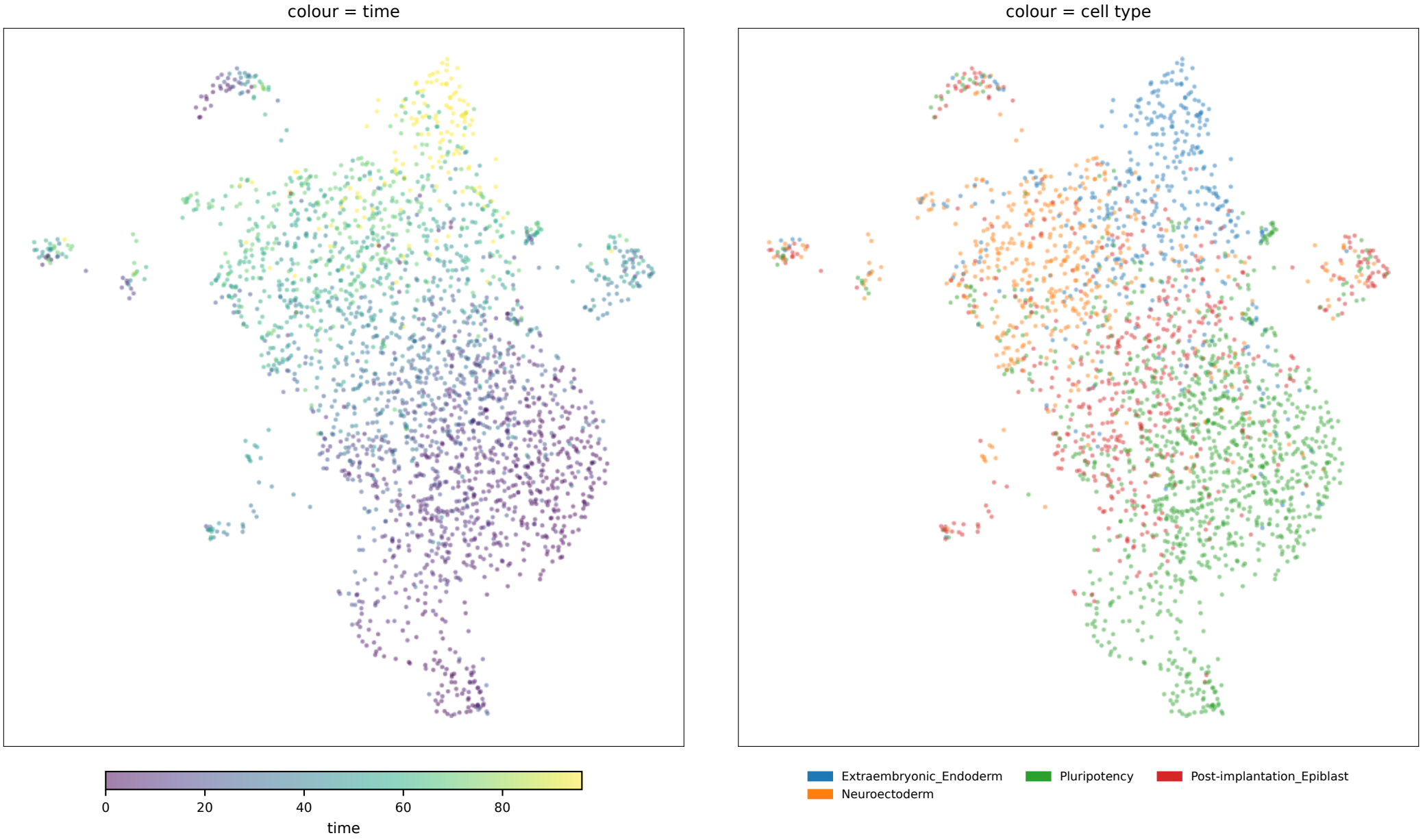
- ✓ P1 KO Dnmt3a
- ✓ P2 OV Dnmt3a

Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

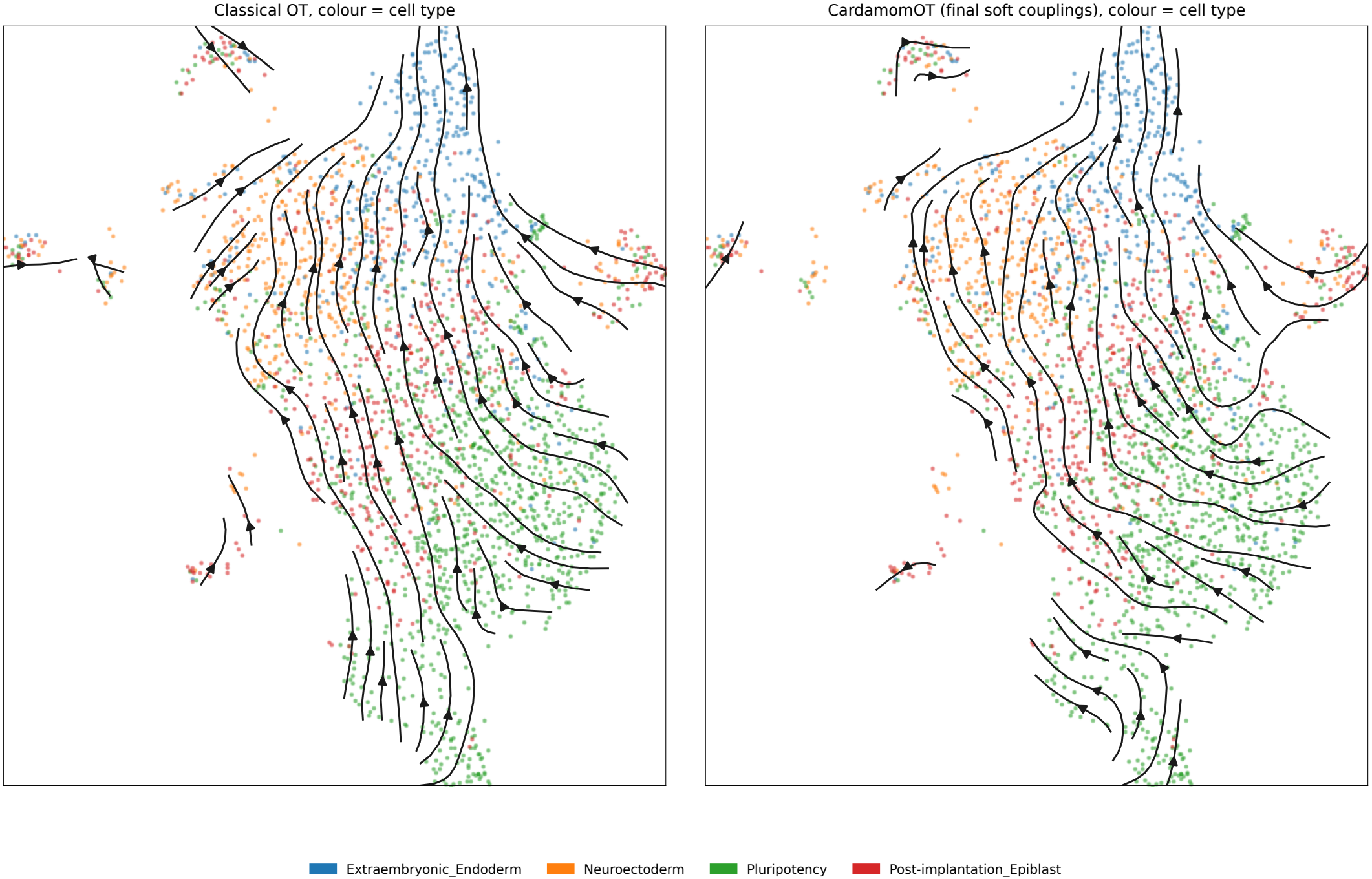
Data — UMAP of all the cells

Data/data.h5ad, 2449 cells: log1p(counts), 41 highly variable genes, z-scored (clipped at ±10), PCA (30 components), kNN graph (20 neighbours), UMAP. Stored in cardamomOT/embedding_data.npz.



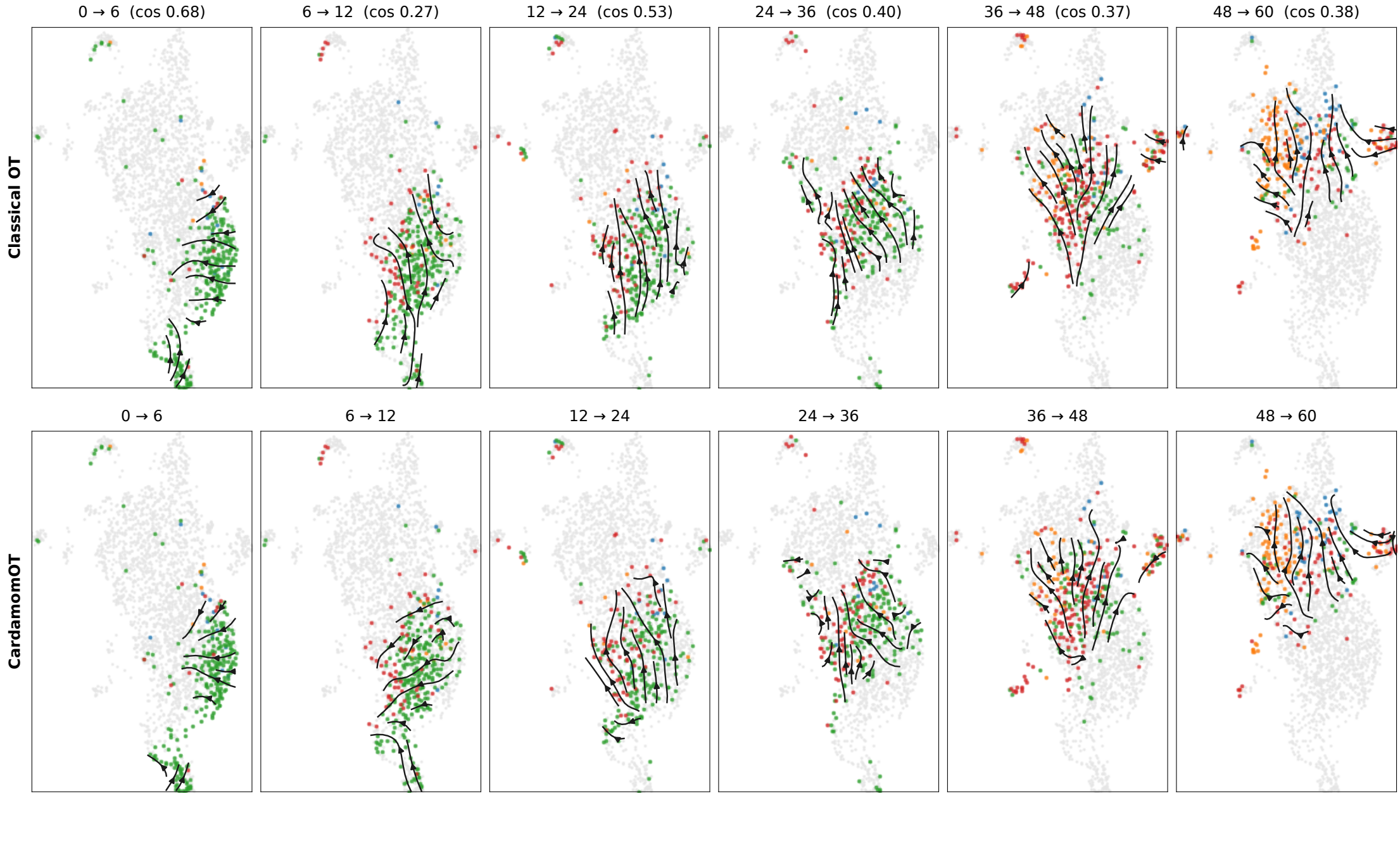
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.50 (PCA space). CardamomOT: final soft couplings of the trajectories (1714 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



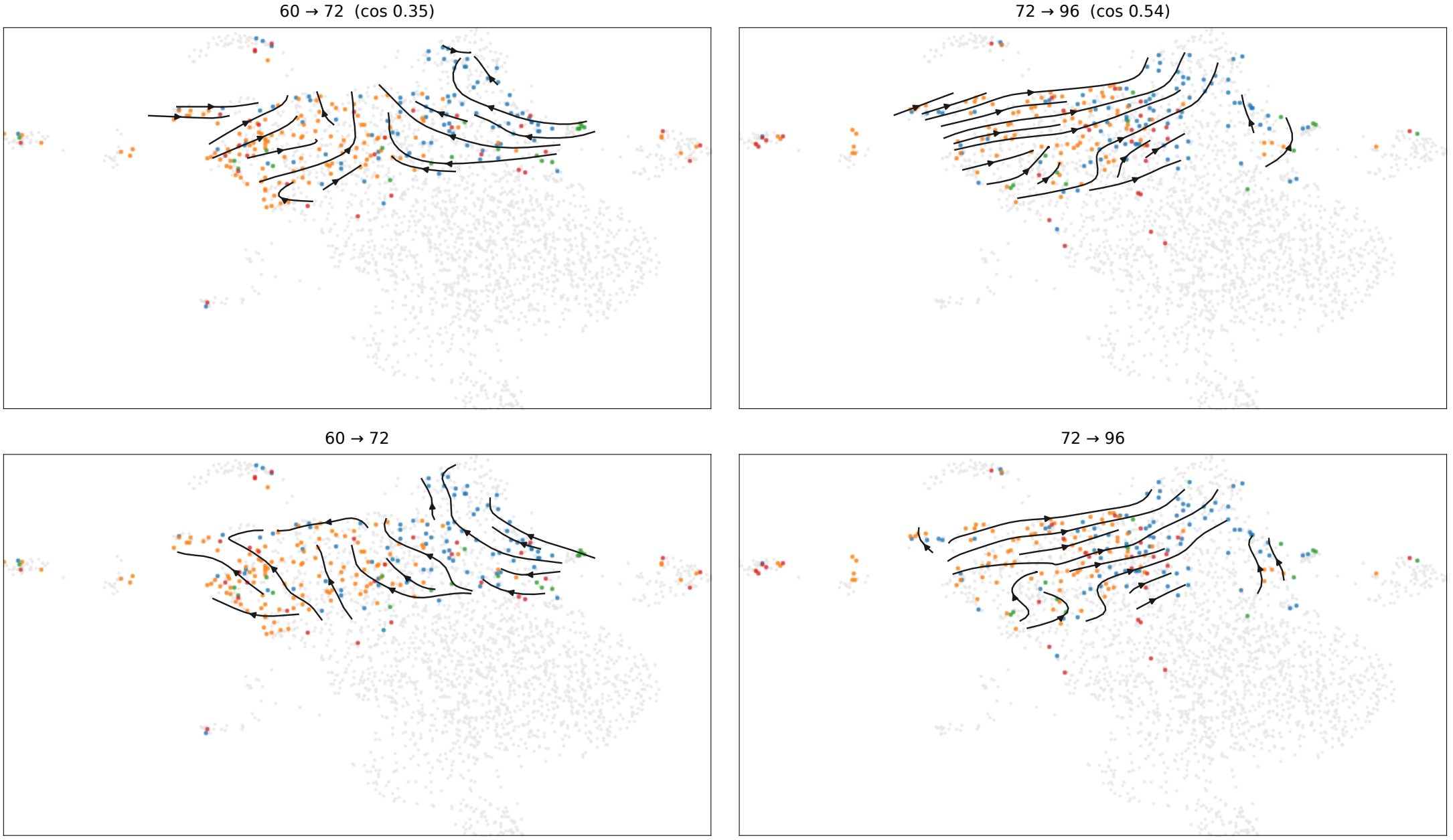
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



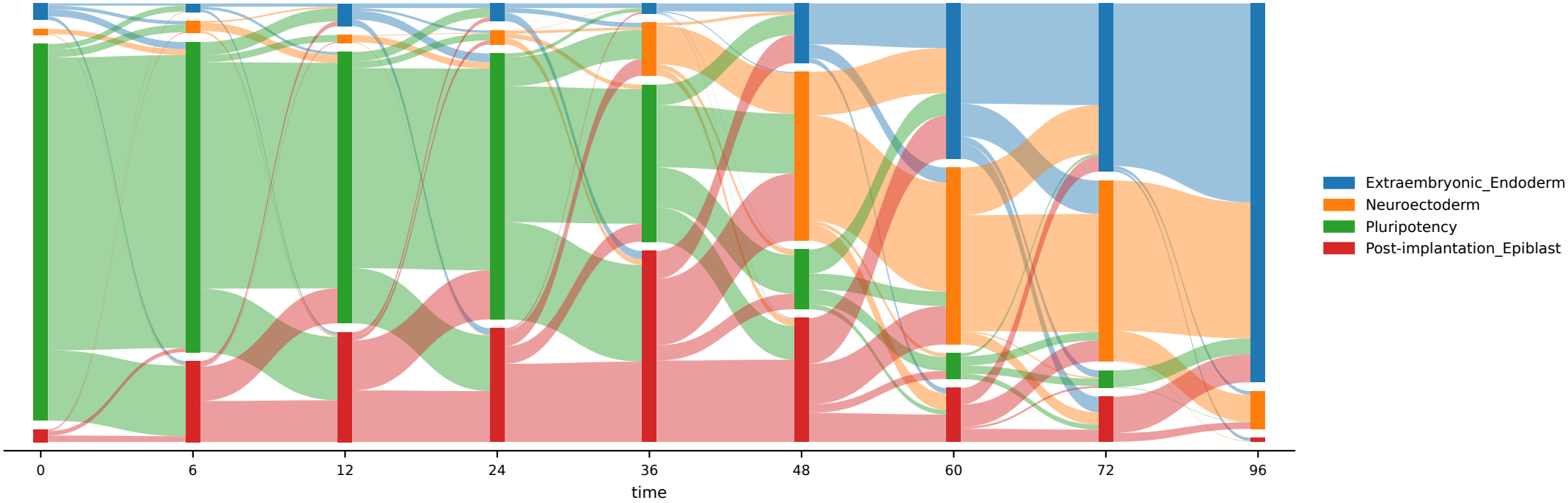
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).

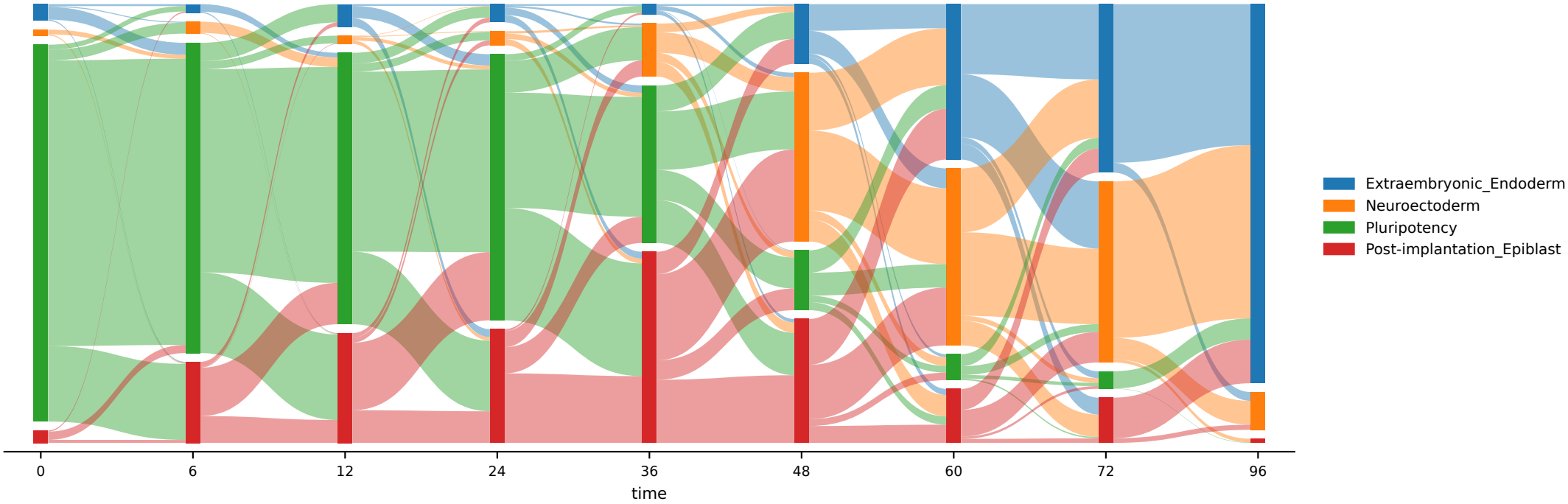


	Top-10 genes up in the ancestors (vs ancestors of the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	**Dab2**	Cd24a	Cdh2	**Krt8**	Meis1	**Gata6**	Hoxb2
Neuroectoderm	**Hoxb2**	**Hoxa1**	**Sox1**	**Sox11**	**Cd24a**	**Pax6**	**Cdh2**	**Meis1**	Dnmt3a	**Nes**
Pluripotency	**Zfp42**	**Esrrb**	**Pou5f1**	**Klf4**	**Klf2**	**Fgf4**	**Sox2**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	Zfp42	**Dppa4**	**Zic3**	**Dnmt3b**	**Dnmt3a**	**Zic2**	**Jarid2**	Klf2	**Pou3f1**

	Top-10 genes up in the cell type (vs the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	**Dab2**	**Gata6**	**Krt8**	Pdgfra	Fst	Sox17	Gata4
Neuroectoderm	**Cd24a**	**Hoxb2**	**Hoxa1**	**Sox1**	**Cdh2**	**Sox11**	**Meis1**	**Pax6**	**Nes**	Sparc
Pluripotency	**Zfp42**	**Pou5f1**	**Esrrb**	**Klf2**	**Fgf4**	**Sox2**	**Klf4**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	**Dnmt3a**	**Dppa4**	**Dnmt3b**	**Zic2**	**Zic3**	**Pou3f1**	Wnt8a	Fgf5	**Jarid2**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Extraembryonic_Endoderm	**Col4a2** Cdh2 **Lamb1** Hoxb2 **Dab2** Cd24a Meis1 **Sparc** Sox11 Pax6	**Sparc** **Lamb1** **Col4a2** **Dab2** Gata6 Krt8 Pdgfra Sox17 Fst Gata4
Neuroectoderm	**Sox1** **Hoxa1** **Hoxb2** **Sox11** Gata6 Dnmt3a **Pax6** Lamb1 Pou3f1 **Meis1**	**Hoxb2** Cd24a **Hoxa1** **Sox1** Cdh2 **Sox11** **Meis1** **Pax6** Nes Sox9
Pluripotency	**Zfp42** **Esrrb** **Klf2** **Fgf4** **Pou5f1** **Klf4** **Sox2** **Nr0b1** **Jarid2** **Dppa3**	**Pou5f1** **Zfp42** **Esrrb** **Klf2** **Fgf4** **Klf4** **Sox2** **Jarid2** **Nr0b1** **Dppa3**
Post-implantation_Epiblast	Pou5f1 Zfp42 **Jarid2** **Dppa2** **Zic3** Esrrb **Dppa4** **Dnmt3b** **Zic2** Klf2	**Dppa2** **Dppa4** Dnmt3a **Dnmt3b** **Zic2** Pou3f1 **Zic3** Wnt8a Fgf5 **Jarid2**

Genes of the model (41)

Data/data_train.h5ad, alphabetical order.

Ascl1
Cd24a
Cdh2
Cer1
Col4a2
Dab2
Dnmt3a
Dnmt3b
Dppa2
Dppa3
Dppa4
Esrrb
Fgf4
Fgf5
Fst
Gata4
Gata6
Hoxa1
Hoxb2
Jarid2
Klf2
Klf4
Krt8
Lamb1
Meis1
Nes
Nr0b1
Pax6
Pdgfra
Pou3f1
Pou5f1
Sox1
Sox11
Sox17
Sox2
Sox9
Sparc
Wnt8a
Zfp42
Zic2
Zic3

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) (depth model 'mean') when the depth factor s_i is used.

Depth

not estimated

Reason

41 genes < 10000

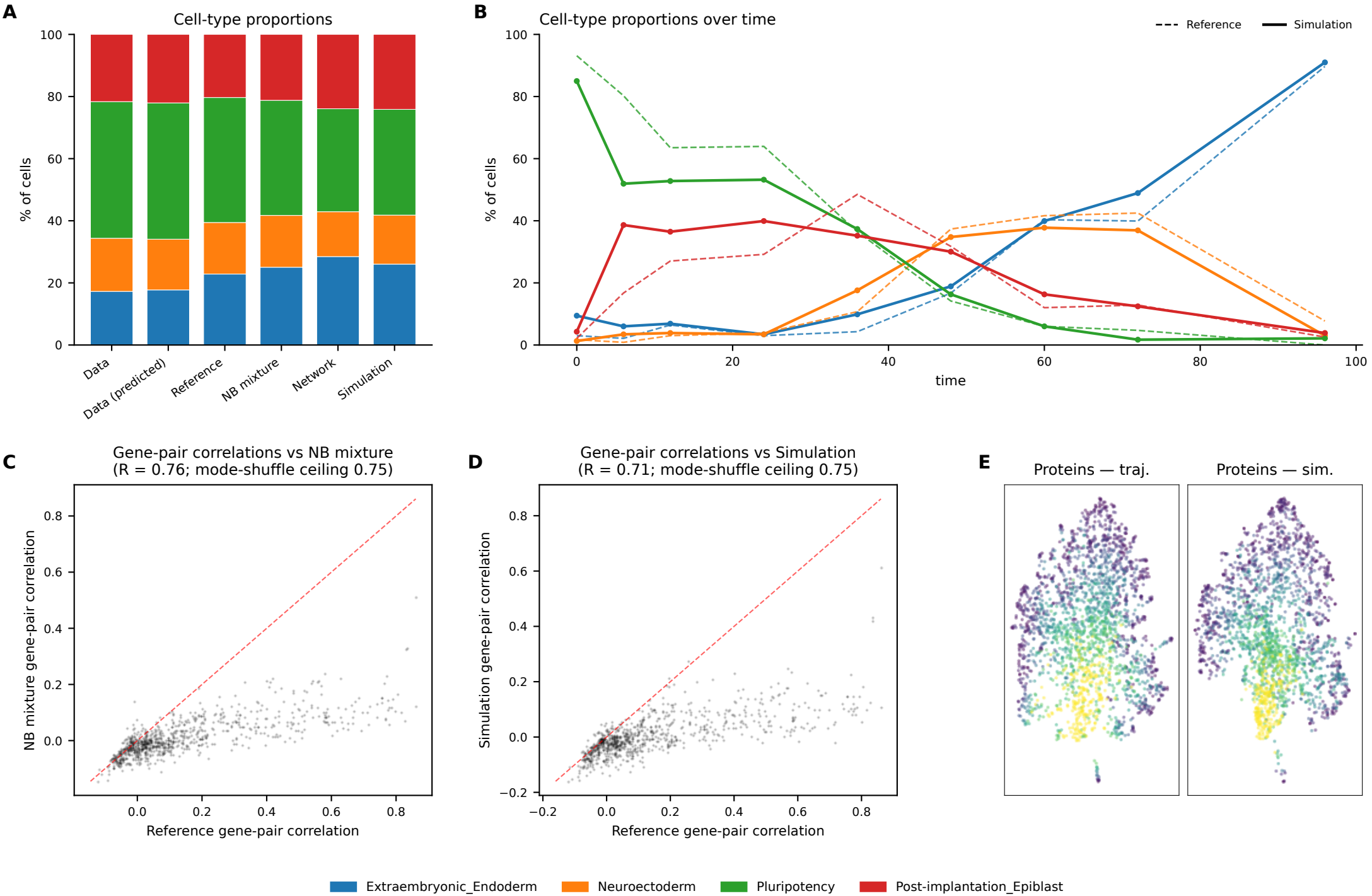
1. Generative model — trajectories and simulation

UMAP learned on the observed data; Reference (states drawn among the observed cells), NB mixture, network modes and simulation (without proliferation) projected onto it. Shuffled reference: each gene of the Reference permuted within (sample, time, mode), the best a model with genes independent given their modes can reach.



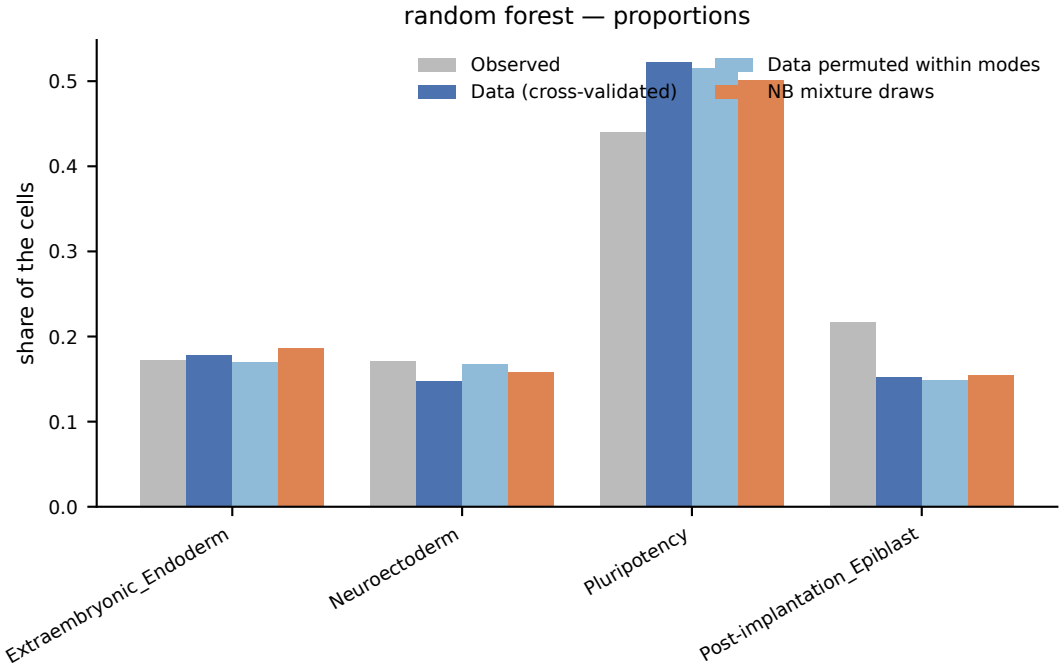
1. Generative model — quantitative checks

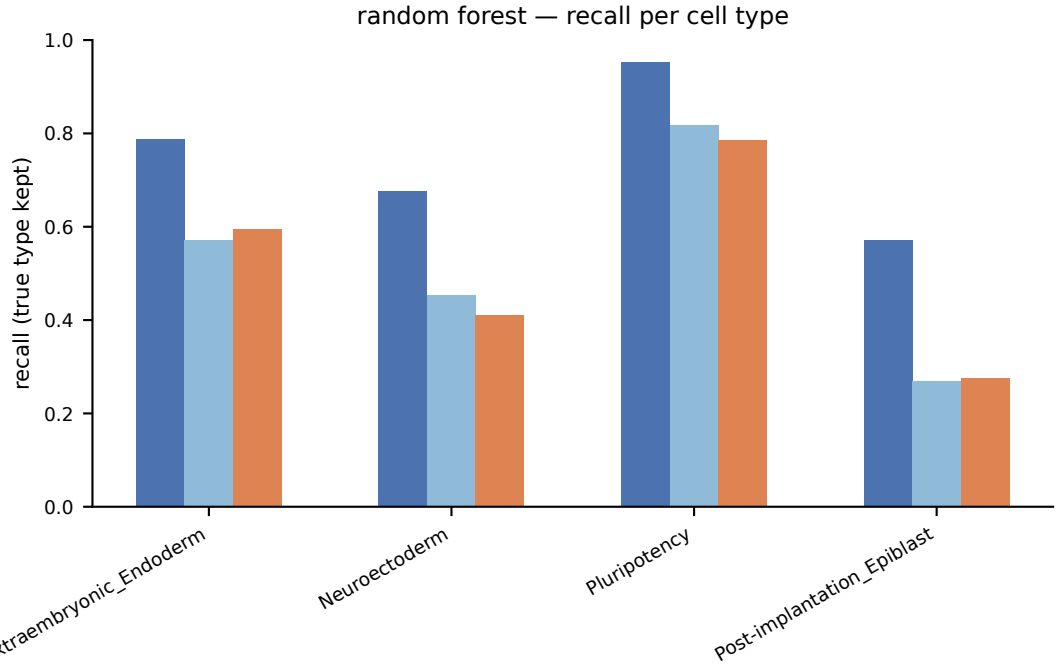
Reference = RNA of the trajectories (states drawn at each time among the observed cells: the observed composition); simulation without proliferation.

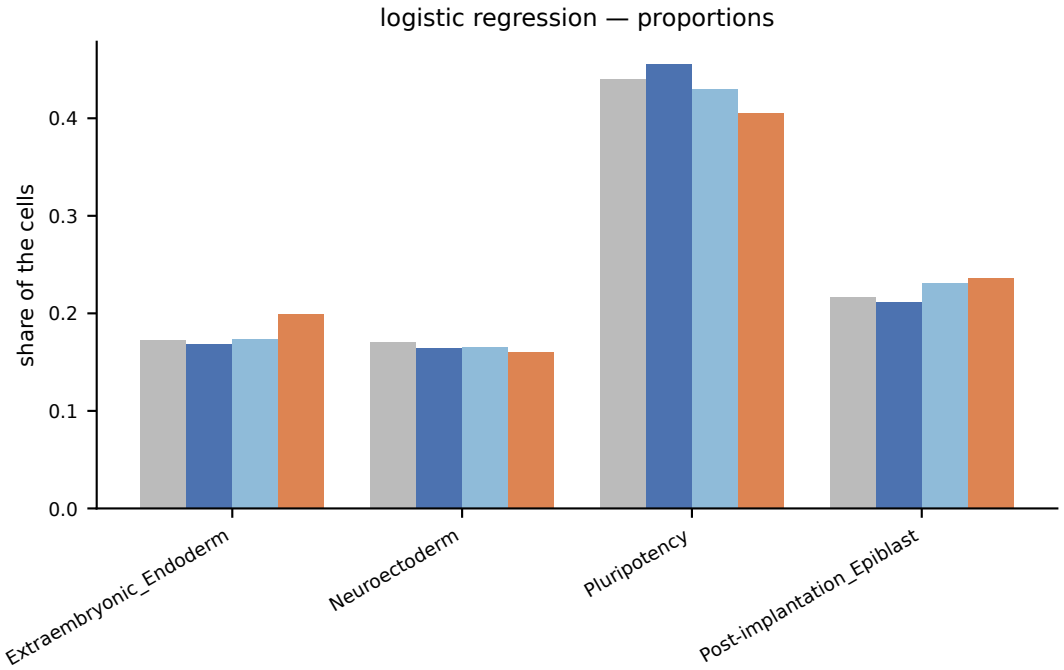


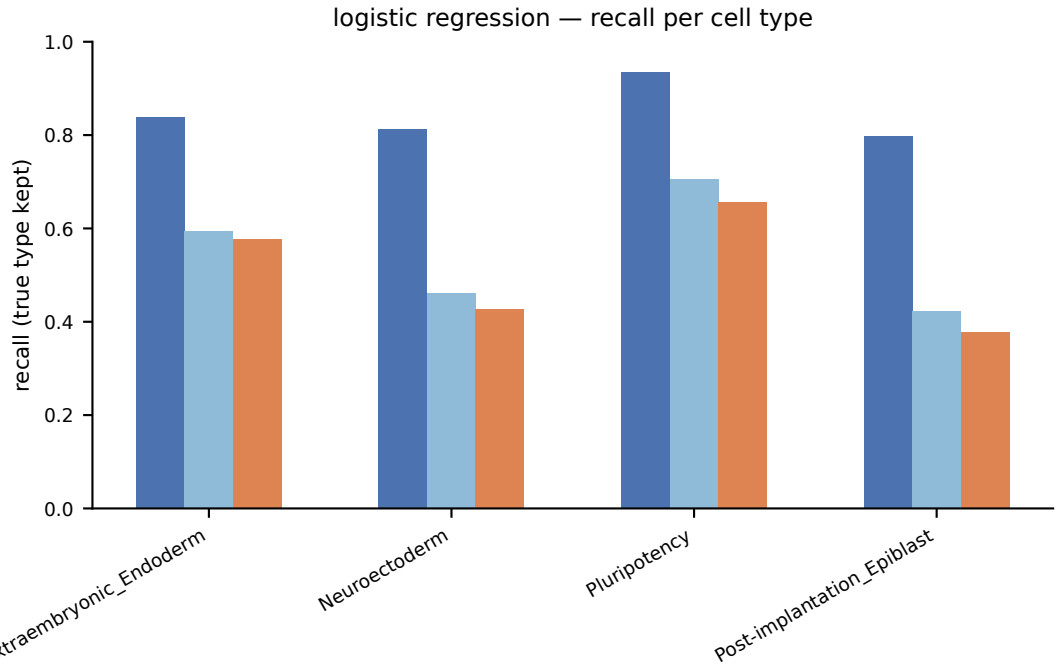
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







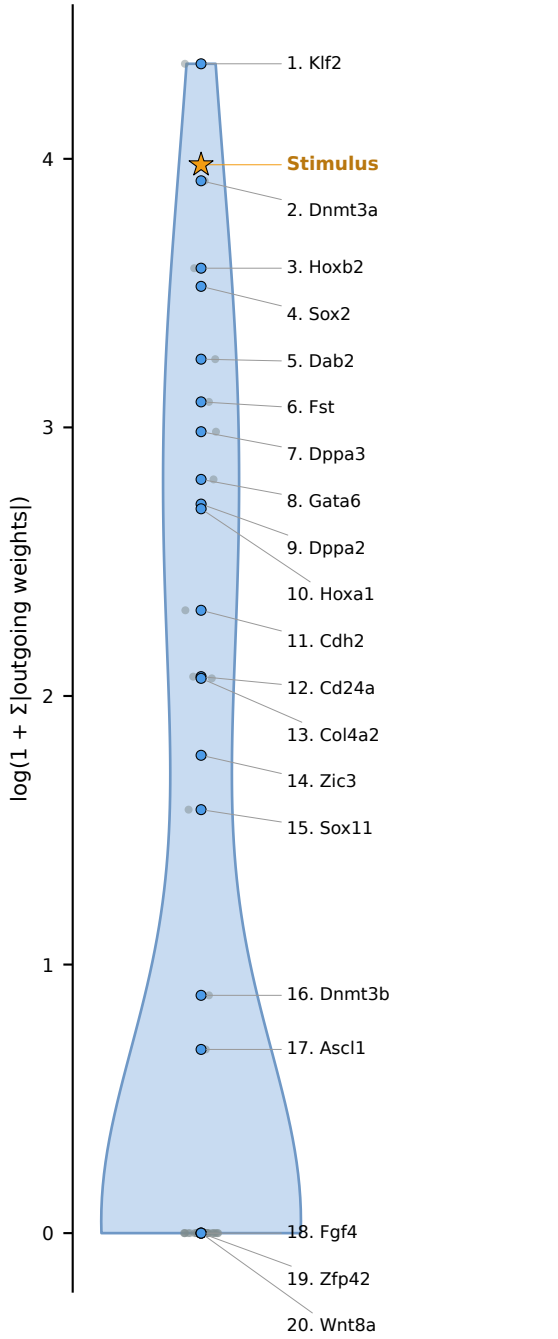


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

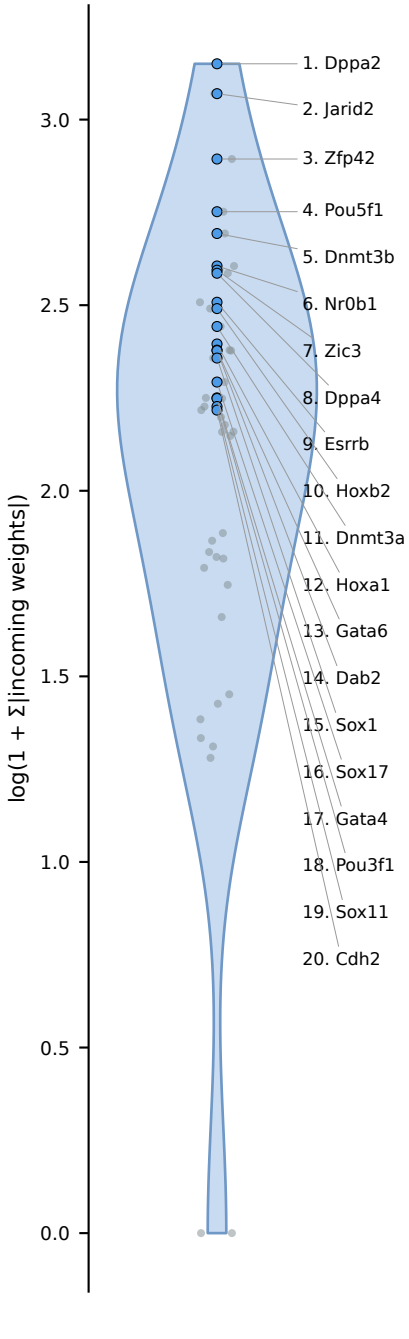
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C Top 20 regulators

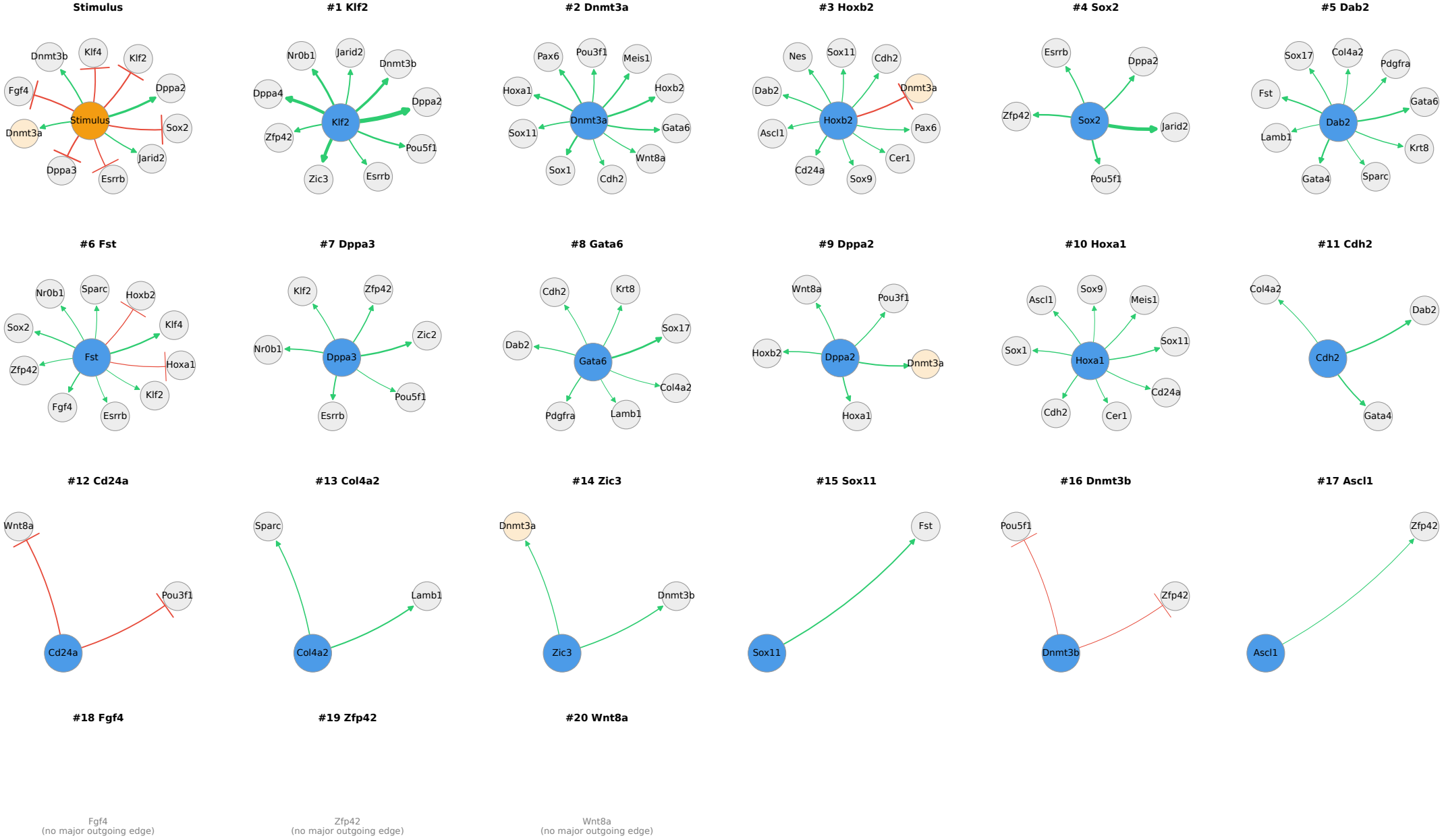
#	Regulator	Σ|w|	act.	inh.
1	Klf2	76.78	9	0
2	Dnmt3a	49.31	13	3
3	Hoxb2	35.34	13	4
4	Sox2	32.96	5	0
5	Dab2	24.90	9	0
6	Fst	21.09	8	3
7	Dppa3	18.77	6	0
8	Gata6	15.55	7	0
9	Dppa2	14.10	5	0
10	Hoxa1	13.83	8	0
11	Cdh2	9.17	3	0
12	Cd24a	6.94	0	2
13	Col4a2	6.89	2	0
14	Zic3	4.92	2	0
15	Sox11	3.84	1	0
16	Dnmt3b	1.42	0	2
17	Ascl1	0.98	1	0
18	Fgf4	0.00	0	0
19	Zfp42	0.00	0	0
20	Wnt8a	0.00	0	0
★	Stimulus	52.41	7	7

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Dppa2	22.34	2	1
2	Jarid2	20.54	2	2
3	Zfp42	17.06	5	2
4	Pou5f1	14.67	3	2
5	Dnmt3b	13.78	2	1
6	Nr0b1	12.54	3	0
7	Zic3	12.39	1	0
8	Dppa4	12.28	1	0
9	Esrrb	11.28	4	0
10	Hoxb2	11.07	2	1
11	Dnmt3a	10.50	2	1
12	Hoxa1	9.97	2	1
13	Gata6	9.79	2	0
14	Dab2	9.78	3	0
15	Sox1	9.57	3	0
16	Sox17	8.90	2	0
17	Gata4	8.49	2	0
18	Pou3f1	8.47	2	1
19	Sox11	8.27	3	0
20	Cdh2	8.18	4	0

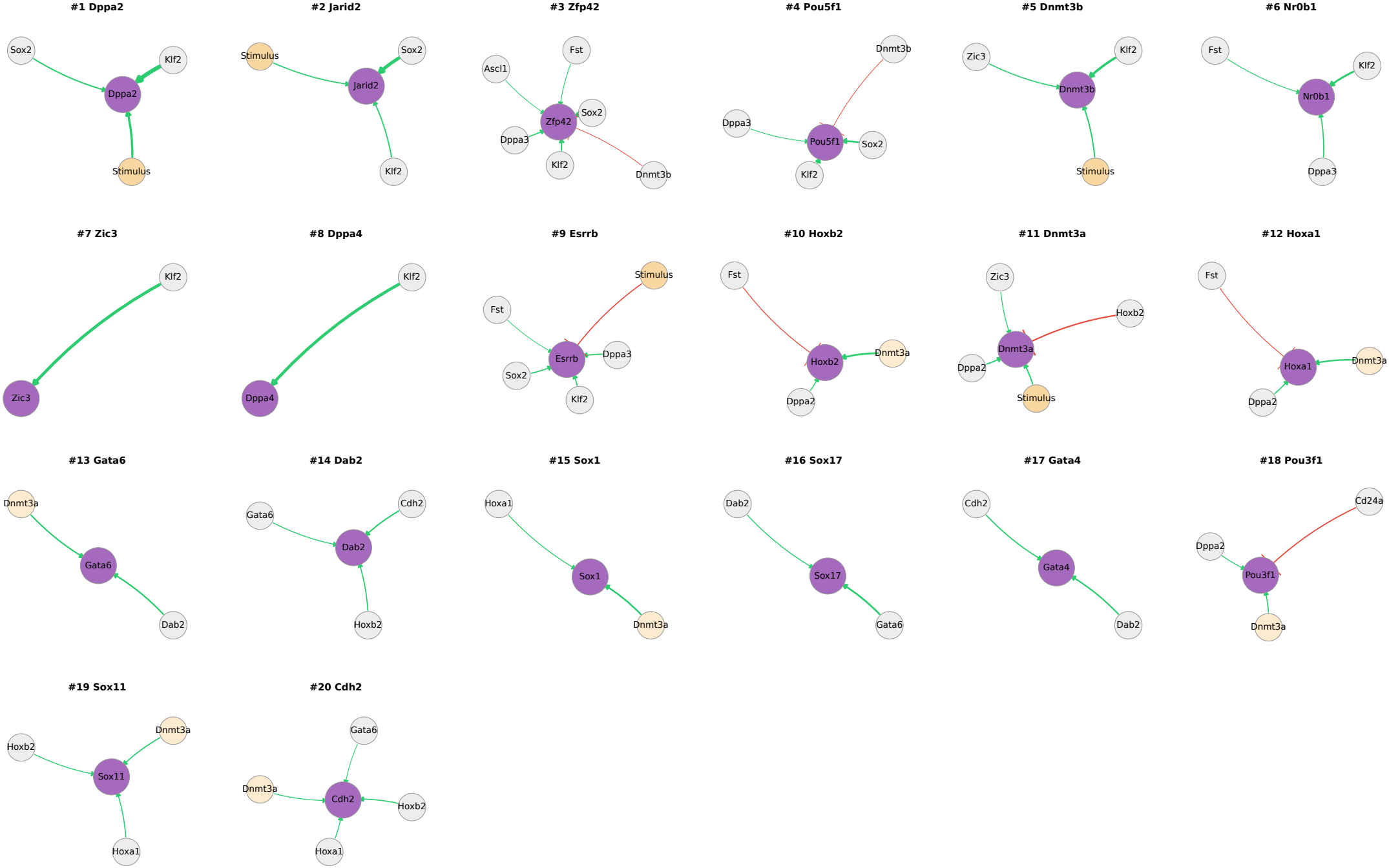
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



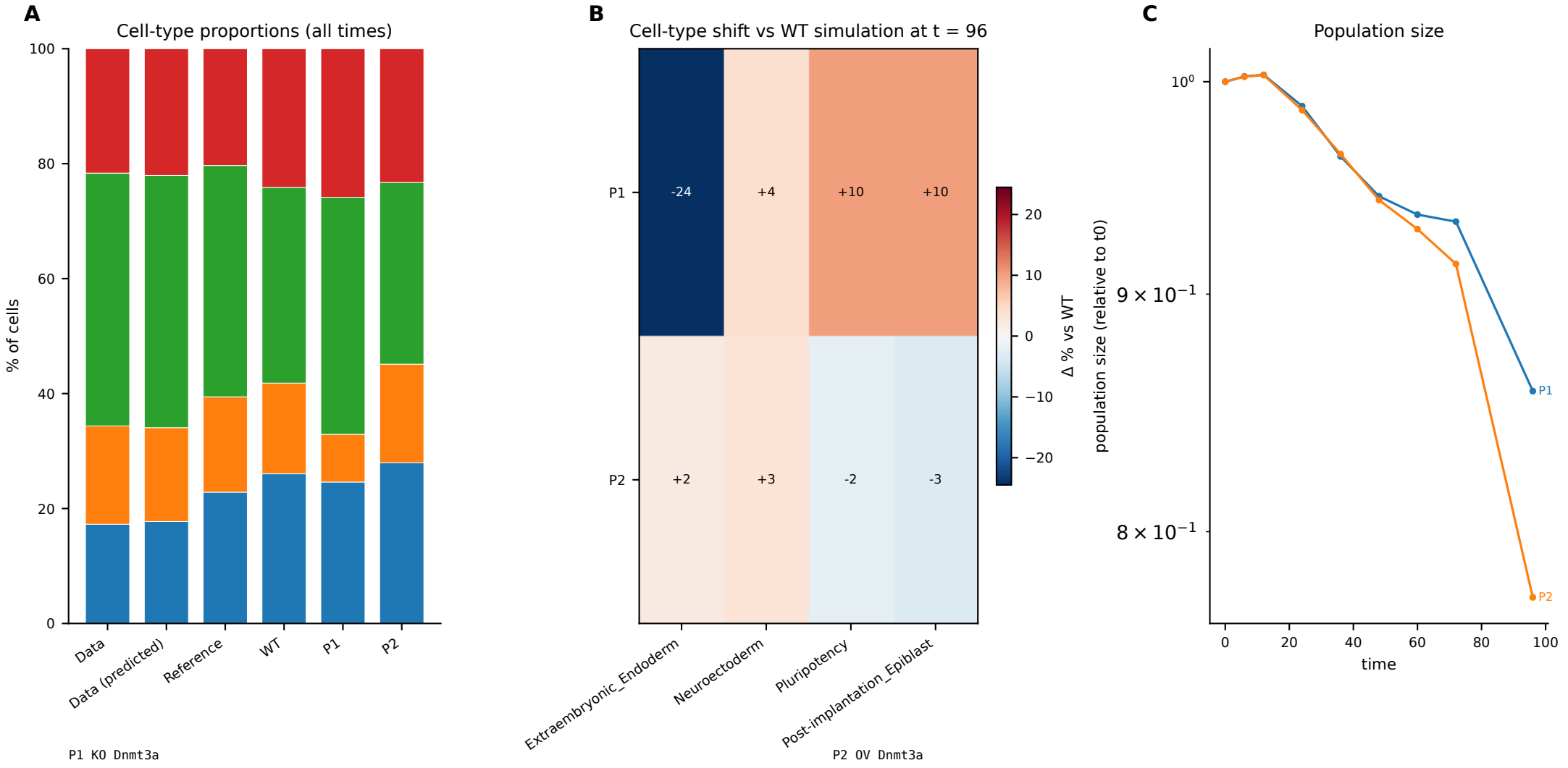
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



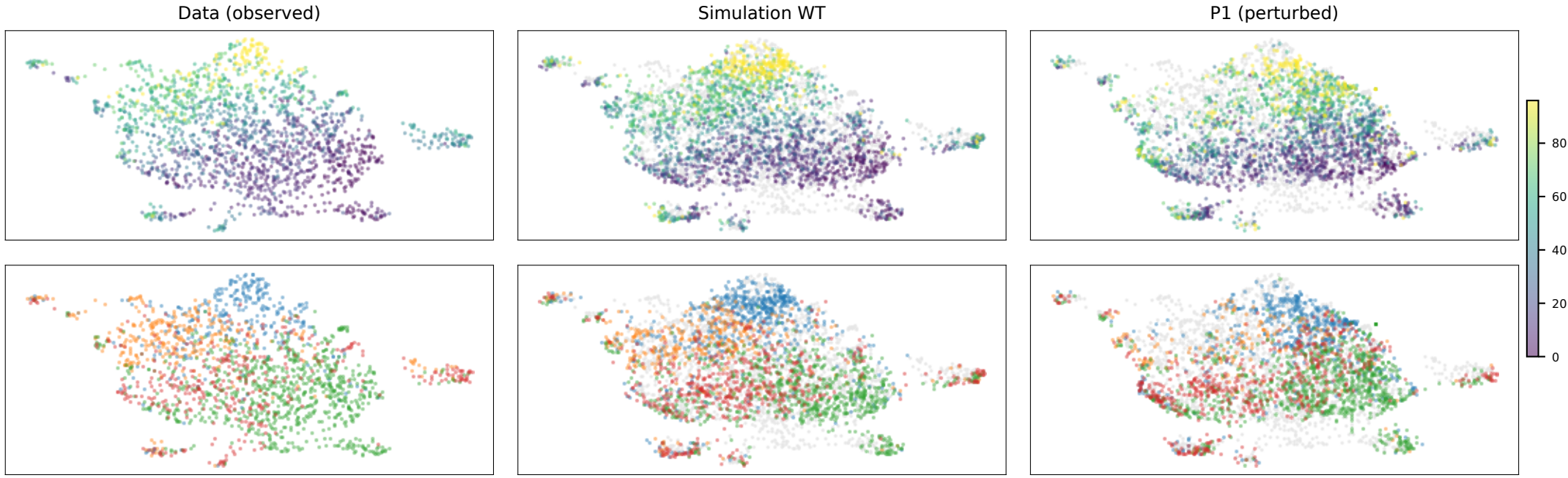
3. In-silico perturbations — overview

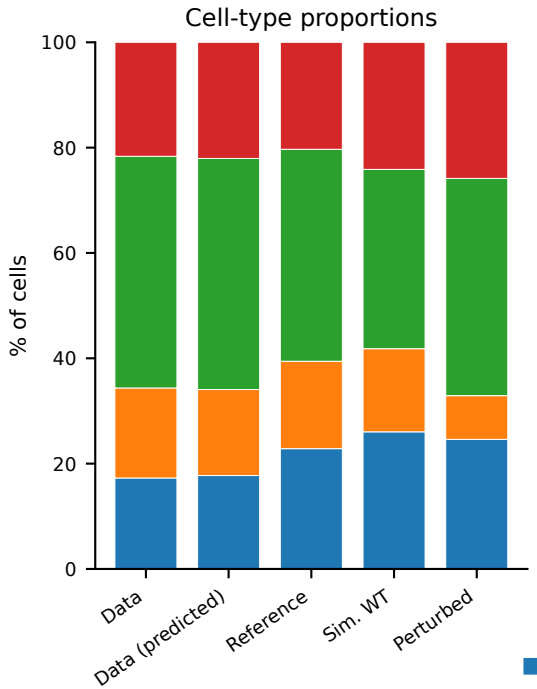
Cell types predicted by per-sample classifiers (logistic, class weights 1 / count^0.5) trained on the observed data. Population sizes from the branching simulation (net rates + RATE effects).

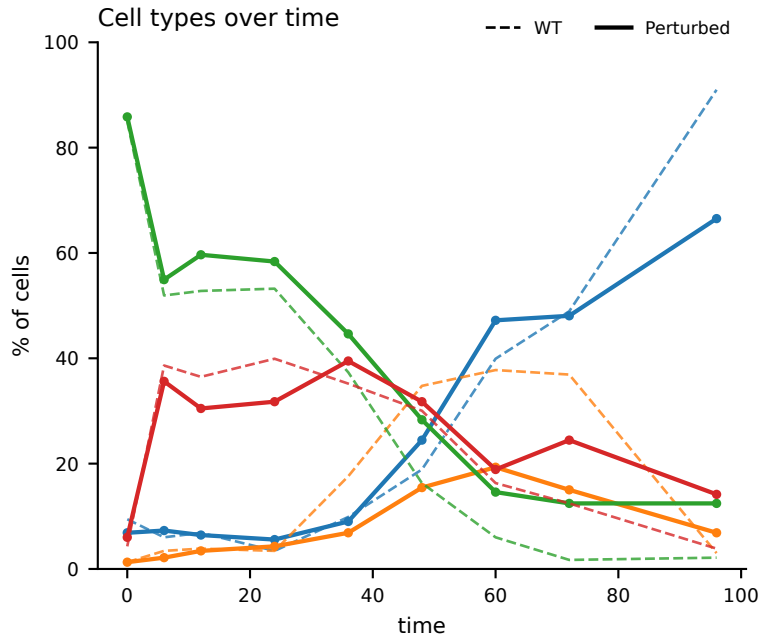


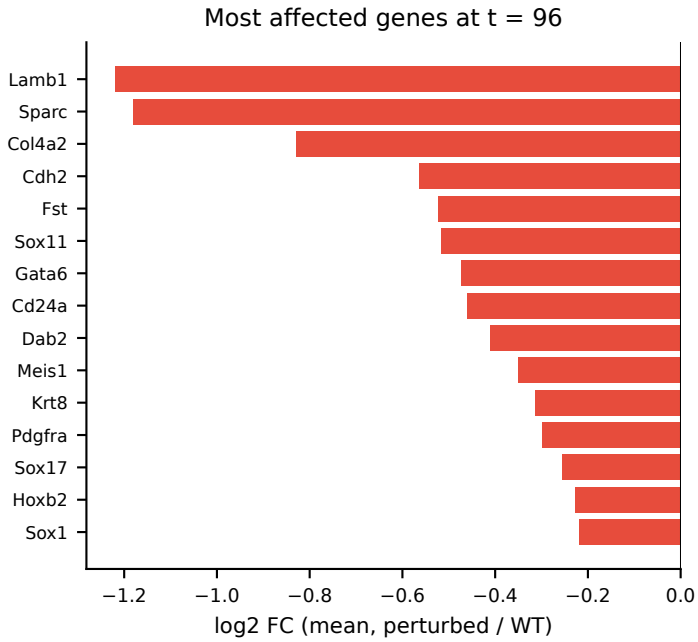
3. Perturbation P1 — KO Dnmt3a

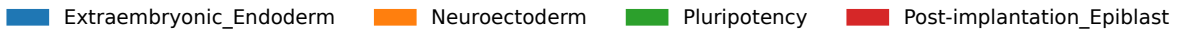
KO_Dnmt3a_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).





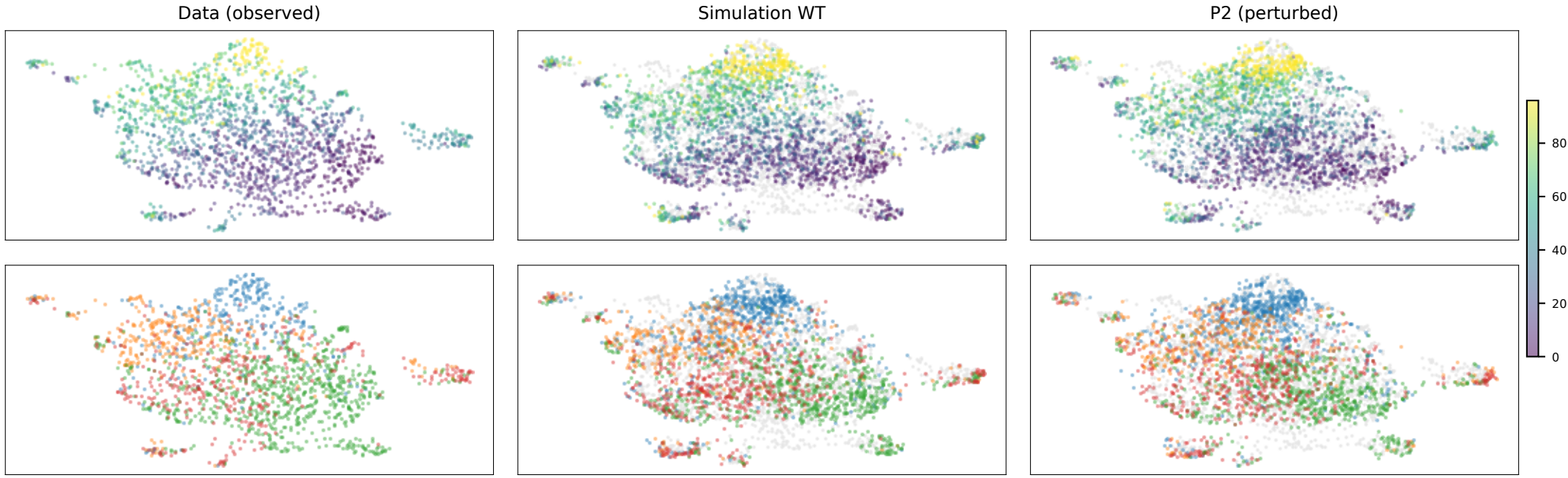


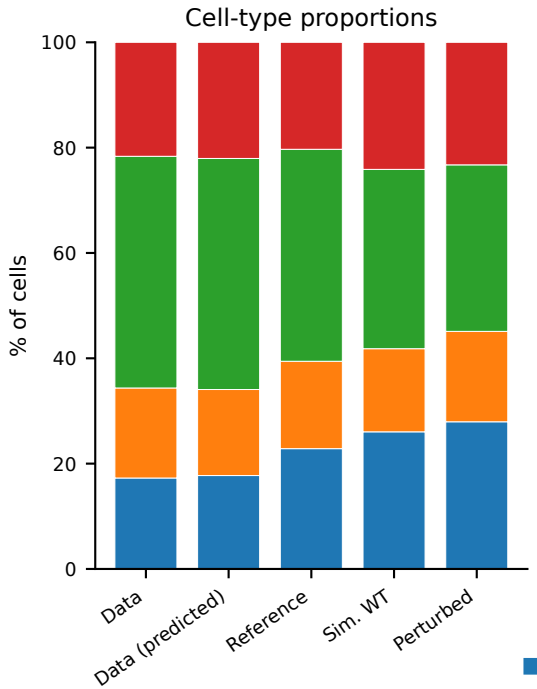


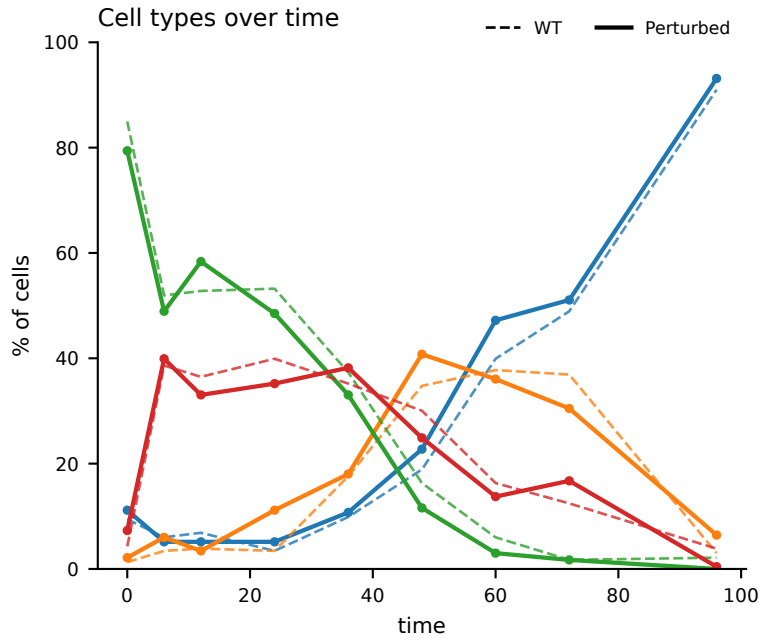


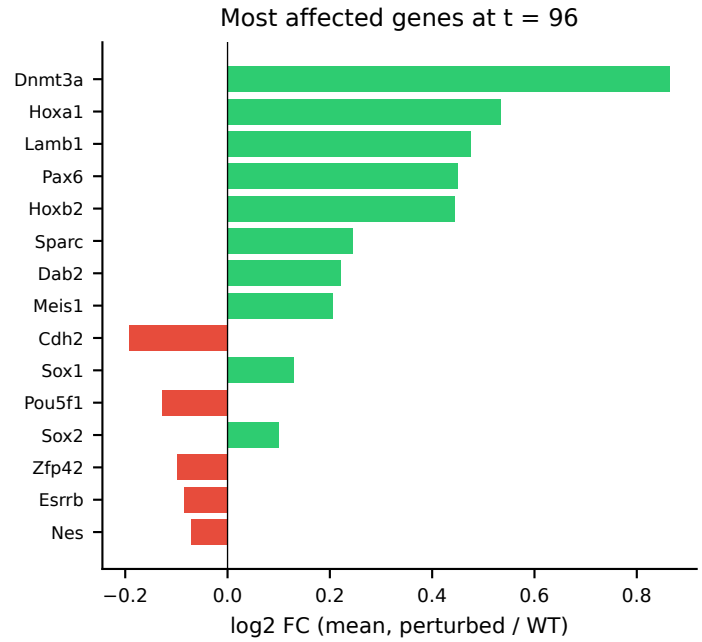
3. Perturbation P2 — OV Dnmt3a

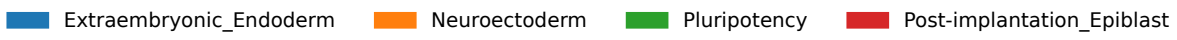
KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).











4. Proliferation

Section unavailable:

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FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
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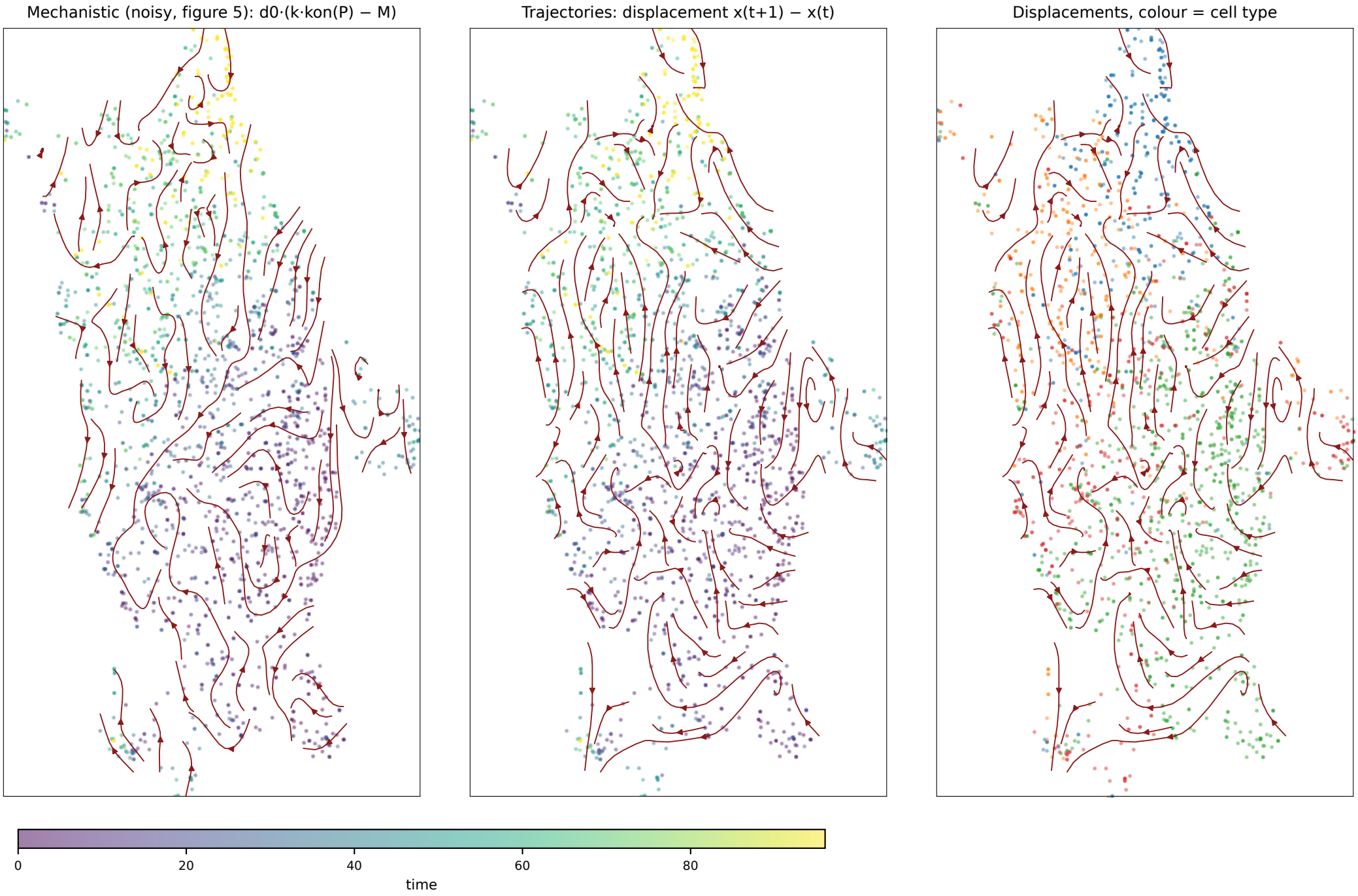
4. Protein dilution

Section unavailable:

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ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
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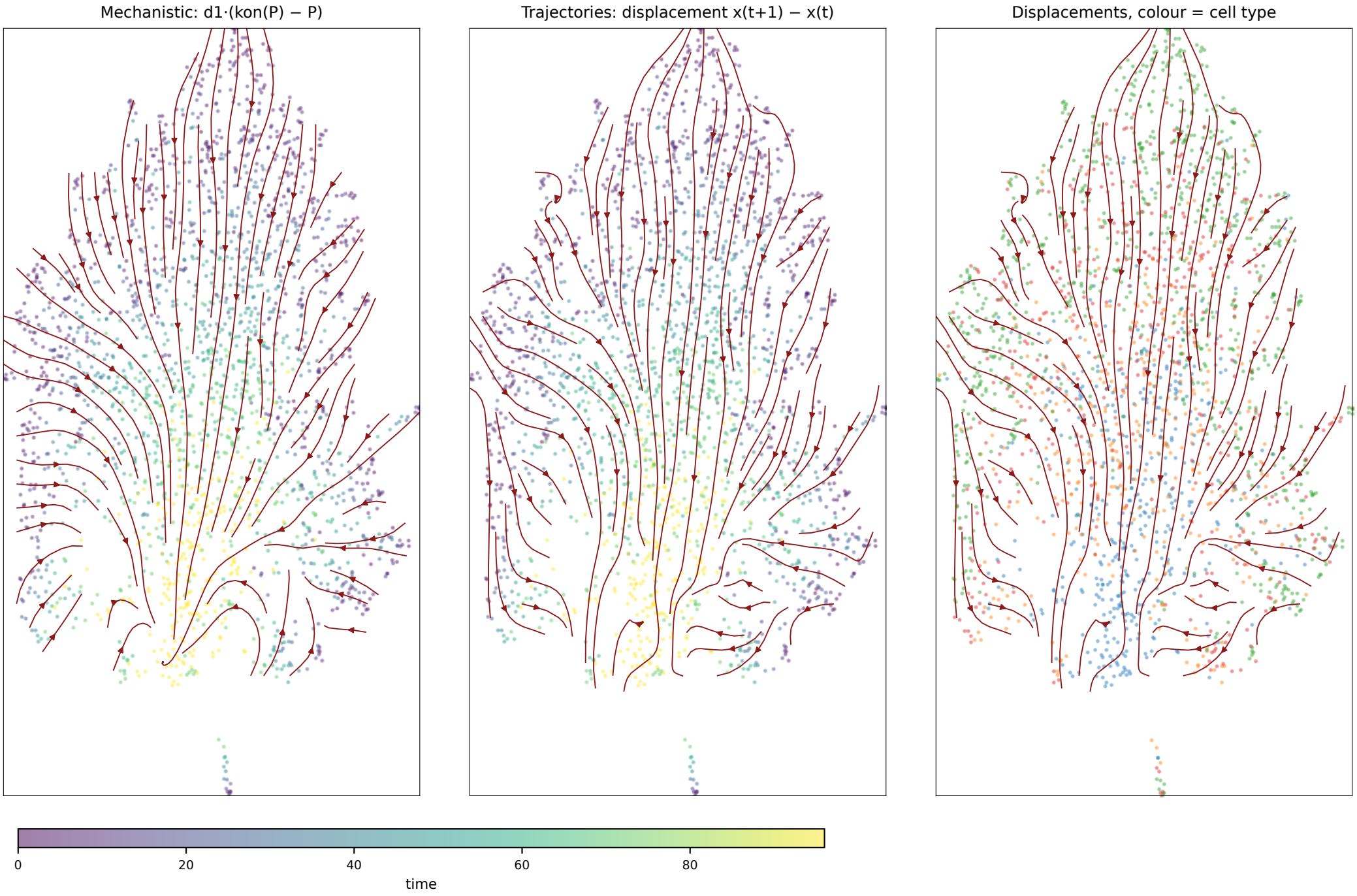
5. Learned dynamics — mRNA velocity and displacement fields

All 2097 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.05.



5. Learned dynamics — Proteins velocity and displacement fields

All 2097 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.13.



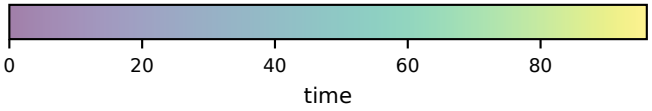
5. Learned dynamics — summary

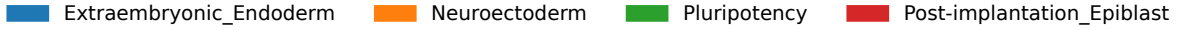
All 2097 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.





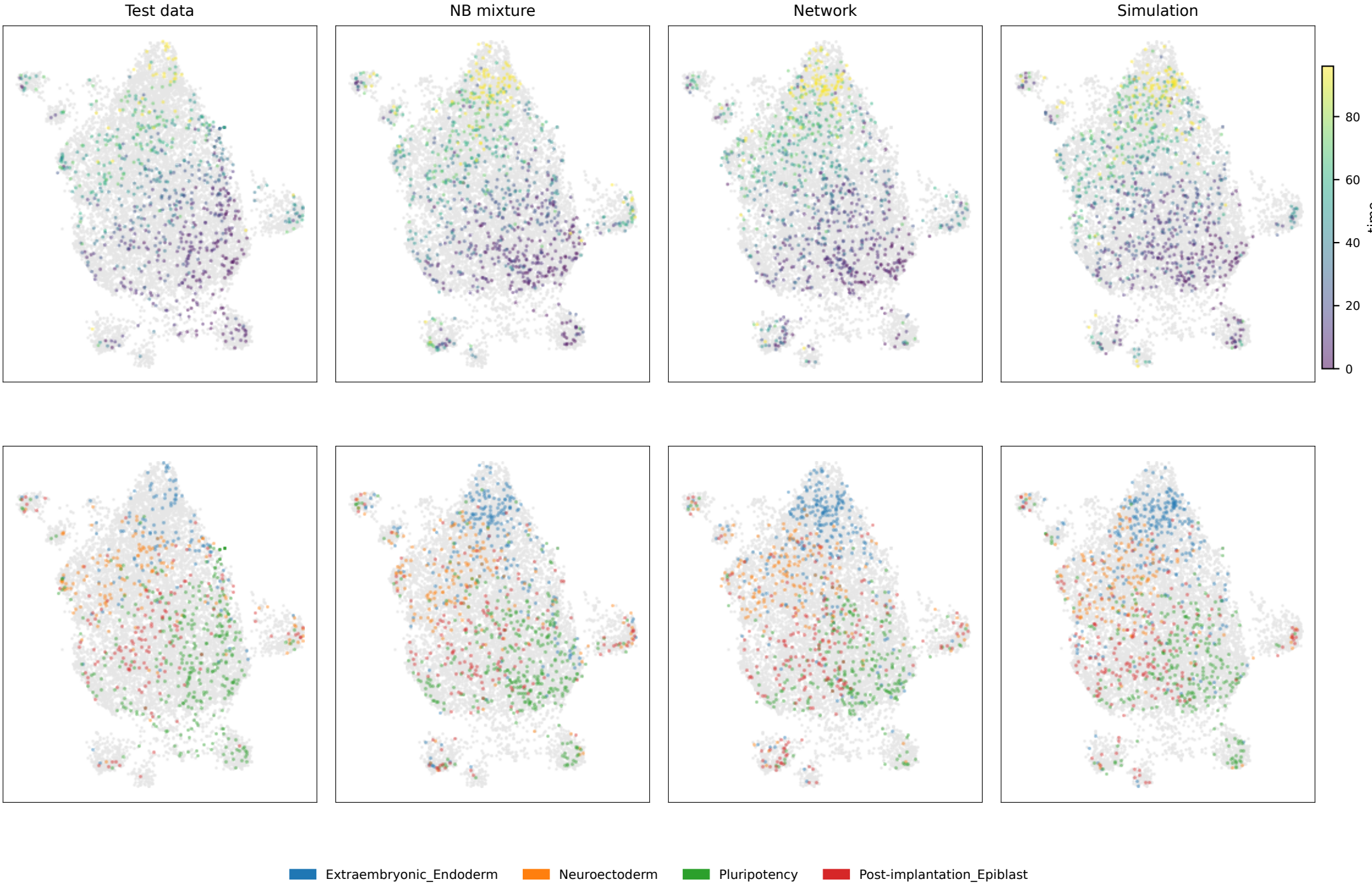
No net proliferation rate





6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

